

CASE STUDY 2

ISMS for Mangalore Refineries & Petrochemicals Ltd.



The Challenge:

MRPL (Mangalore Refineries & Petrochemicals Limited), operates one of the largest refineries in the world, in Mangalore, a maritime city in the south-western part of India. The biggest concern security challenge for the security establishment at MRPL is the nearly 18 km long perimeter, which snakes through difficult and inhospitable terrain. The MRPL refinery itself is also located almost in a forest. Petty thefts are a regular occurrence with thieves breaching the perimeter for stealing valuable items. Of course, it is also a property of national and strategic importance and hence always under the radar of terrorists. The security of the entire premises is handled by the CISF (Central Industrial Security Force).

The Solution:

After undertaking a detailed risk profiling of the entire geography surrounding the Mangalore refinery, MIPL designed a technological solution that involves:

- Perimeter intrusion detection technology (electric fence cant be used due to proximity to village population) supplemented by PTZ cameras which pan to an intrusion location in case of an intrusion attempt
- A network of 220 high resolution fixed and panoramic day & night cameras.
- A GPON-based communication network to minimise points of failure at the field level.
- Redundancy at multiple levels – control room, network, field elements & electricity.
- Biometric Access Control
- Vehicle Tracking System for emergency vehicles
- Two mirrored control rooms
- Monitoring of remote MRPL locations

The project has been successfully completed in December 2018. **It is the first instance of a deployment of a Command & Control solution in a refinery in India and also the first installation of a GPON based dedicated security network in the oil & gas industry in**

India. The project received the Shield Award in 2019, for the best security project in an industrial premises.